STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 2: Motion in a Straight Line

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define instantaneous velocity.
2. Slope of an x-t graph?
3. Acceleration at constant velocity?
4. Can speed be negative?
5. SI unit of acceleration?
6. Runner returns to start: net displacement?
7. Equation linking v,u,a,t?
8. Area under v-t graph?

Section B 2-mark questions - 6 x 2 = 12 marks

9. m in 30 s: average speed?
10. Velocity 4 to 16 m/s in 6 s: acceleration?
11. m/s car brakes at 2 m/s^2: stopping time?
12. Convert 90 km/h to m/s.
13. Move 8 m east then 3 m west: distance/displacement?
14. From rest at 3 m/s^2 for 4 s: final speed?

Section C 3-mark questions - 4 x 3 = 12 marks

15. u=6 m/s, a=2 m/s^2, t=5 s: find v,s.
16. Train at 20 m/s stops in 5 s: a and distance?
17. Velocity rises 2 to 14 m/s in 6 s: a and displacement?
18. u=12 m/s, a=1.5 m/s^2 for 8 s: v,s?

Section D 4-mark questions - 2 x 4 = 8 marks

19. Car at 25 m/s brakes at 5 m/s^2: stop time and distance?
20. x=4+3t-t^2: v(t), a and time at rest?